SpaceX Falcon 9
First-Stage Landing Prediction

Data Science Capstone Project Report

Presented by: [Your Name] | 2026-08-02

OVERVIEW

Executive Summary

Methods used: Data was collected via the SpaceX REST API and Wikipedia web scraping, cleaned and labeled, explored with visualization and SQL, mapped geospatially with Folium, and modeled with four classifiers (Logistic Regression, SVM, Decision Tree, KNN) tuned via GridSearchCV.

Key results: Across 90 Falcon 9 launches, the overall first-stage landing success rate was 67%. The best-performing model, Logistic Regression, reached 83% test accuracy in predicting landing outcome.

90	**67%**	**83.3%**	**4**
Launches analyzed	Overall landing success rate	Best model test accuracy	ML models compared

Conclusion: Landing success is highly predictable from launch parameters, meaning a competing launch provider could reliably estimate SpaceX's true reusable-launch cost advantage.

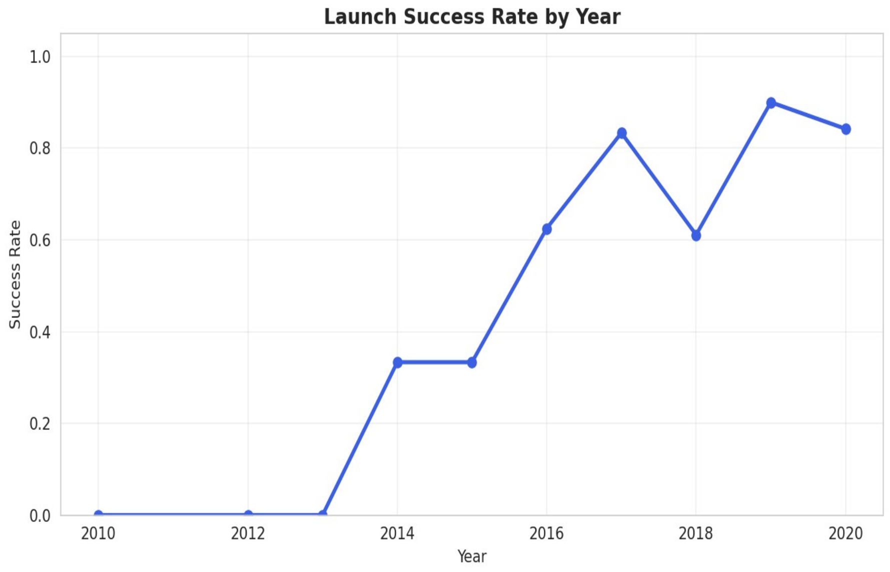
Introduction

SpaceX advertises Falcon 9 launches at $62 million, versus $165M+ from other providers — largely because SpaceX reuses the rocket's first stage instead of discarding it.

Problem statement: If the first stage does not land successfully, that launch's cost estimate changes substantially. This project builds a model to predict whether the first stage will land, using historical launch data — enabling accurate cost estimation for a competing bid.

Objectives:

- Collect and clean historical Falcon 9 launch data
- Identify factors correlated with landing success
- Visualize patterns across sites, orbits, and time
- Build and evaluate a landing-outcome classifier



Falcon 9 landing success climbed steadily as SpaceX matured its recovery process.

Data Collection – SpaceX API

Pipeline:

1. GET /v4/launches/past → all historical launches
2. Filter to single-core, single-payload Falcon 9 flights
3. For each launch, call /rockets, /launchpads, /payloads, /cores endpoints
4. Assemble into one row per launch: booster, orbit, site, outcome, mass

`api.spacexdata.com/v4/launches/past`

Result: 90 clean Falcon 9 launch records (2010–Nov 2020), each with booster version, payload mass, orbit, launch site, flight count, grid fins, reused flag, legs, landing pad, and outcome.

Falcon 1 launches were excluded — this project focuses only on the Falcon 9 landing-prediction problem.

GitHub: https://github.com/YOUR-USERNAME/spacex-falcon9-capstone/blob/main/notebooks/01_data_collection_api.py

Data Collection – Web Scraping

Pipeline:

1. Request the Wikipedia "List of Falcon 9 and Falcon Heavy launches" page
2. Parse HTML tables with BeautifulSoup, locate the wikitable elements
3. Extract column headers, then walk each row of launch data
4. Clean text (dates, booster serials, payload mass) into a dataframe

```
BeautifulSoup(response.text, 'html.parser')
```

This gives an independent, cross-checked source for launch records — used to validate the API dataset and fill in fields (like precise payload customer) the API doesn't expose.

Key techniques: regex/Unicode cleanup for payload mass strings, and helper functions to pull booster version and landing status out of merged table cells.

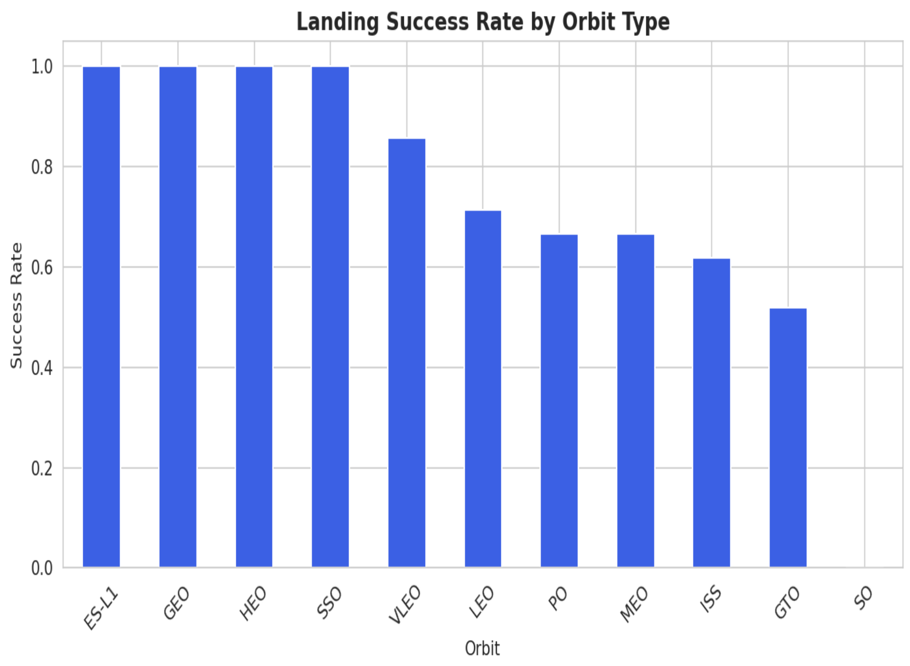
GitHub: https://github.com/YOUR-USERNAME/spacex-falcon9-capstone/blob/main/notebooks/02_web_scraping.py

Data Wrangling Methodology

Missing values: PayloadMass (5.6% missing) filled with column mean; LandingPad (28.9% missing) kept as-is since a null pad legitimately means "no ground/drone-ship landing occurred."

Label engineering: The raw Outcome field (e.g. "True ASDS", "False Ocean", "None None") was converted into a binary Class label — 1 if the booster landed successfully by any method, 0 otherwise.

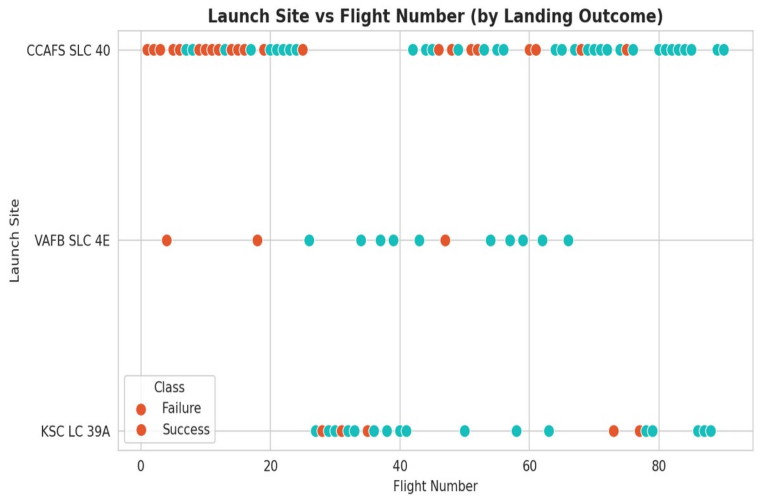
Result: 90 launches → 60 successful landings (66.7%), 30 unsuccessful (33.3%)



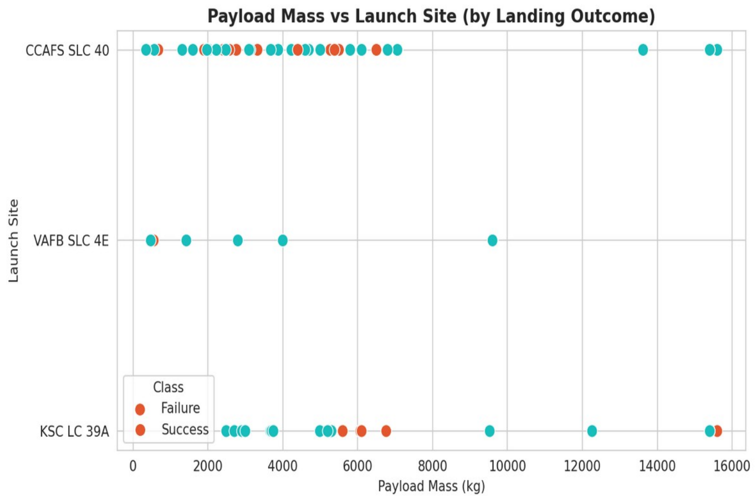
GitHub: https://github.com/YOUR-USERNAME/spacex-falcon9-capstone/blob/main/notebooks/03_data_wrangling.py

EDA with Data Visualization

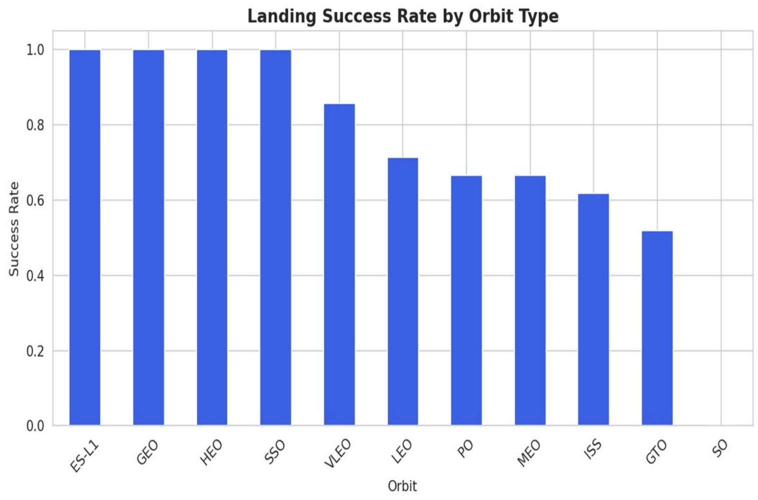
We plotted flight number, payload mass, orbit type, and launch year against landing outcome to surface which factors correlate with success.



Flight # vs Launch Site



Payload vs Launch Site

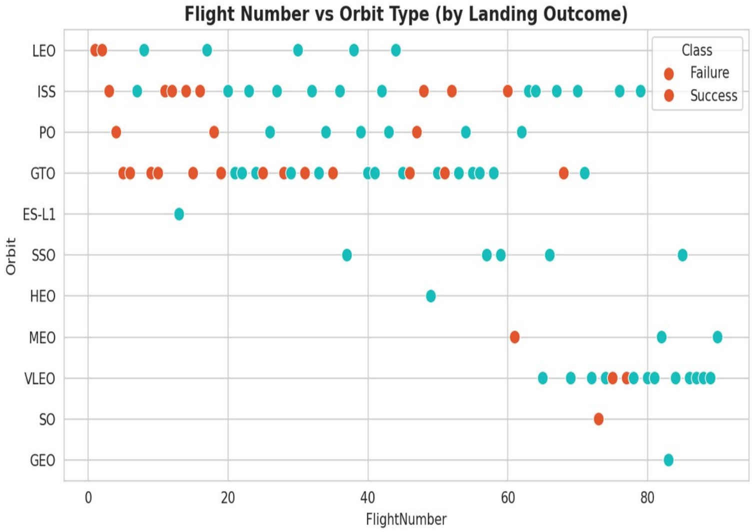


Success Rate by Orbit

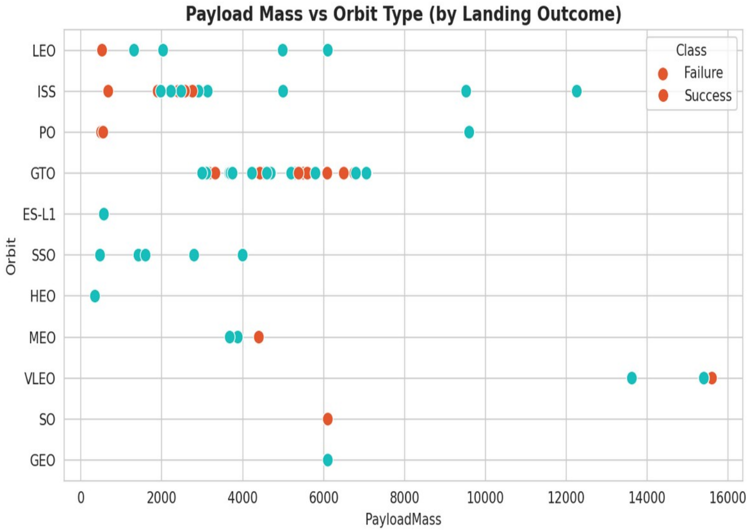
Chart types used: scatter plots (flight #/payload vs site or orbit), bar charts (success rate by orbit), and a line chart (yearly trend, see Introduction slide).

GitHub: https://github.com/YOUR-USERNAME/spacex-falcon9-capstone/blob/main/notebooks/04_07_eda_sql_ml.py

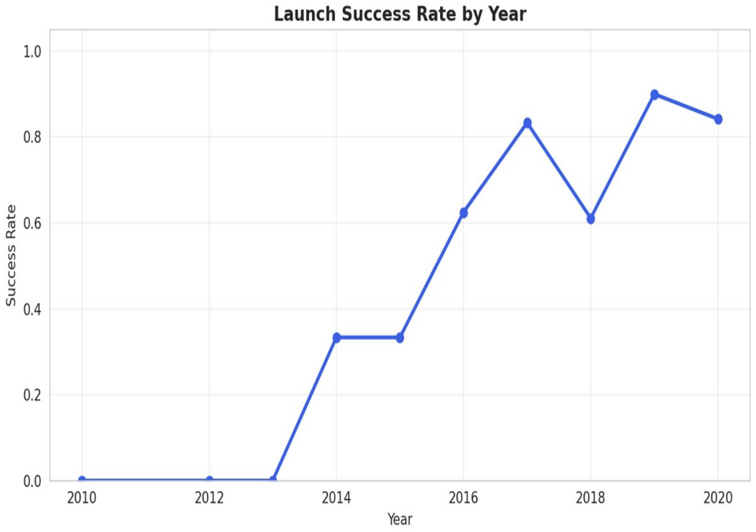
EDA Visualization — Orbit & Trend Detail



Flight # vs Orbit Type



Payload vs Orbit Type



Success Rate by Year

Takeaway: GTO and ISS-orbit missions (heavier, higher-energy) show lower success rates; success climbed year-over-year as SpaceX refined the recovery process, from ~0% in 2010–2013 to consistently near 100% by 2019–2020.

EDA with SQL

The scraped launch table was loaded into a SQL database (SQLite / IBM Db2) as SPACEXTABLE, then queried directly to answer specific business questions.

- ✓ Distinct launch sites used
- ✓ Total payload mass carried for NASA (CRS)
- ✓ Average payload mass for F9 v1.1 boosters
- ✓ Date of the first successful ground-pad landing
- ✓ Boosters that succeeded on a drone ship with 4,000–6,000 kg payload
- ✓ Count of each mission outcome (success/failure)
- ✓ Booster(s) carrying the maximum payload mass
- ✓ Failed drone-ship landings in 2015
- ✓ Ranking of landing outcomes between 2010–2017

GitHub: https://github.com/YOUR-USERNAME/spacex-falcon9-capstone/blob/main/notebooks/04_07_eda_sql_ml.py

EDA SQL Results — Sites & Payload

Launch Sites (4 distinct)

CCAFS LC-40 · CCAFS SLC-40 · KSC LC-39A · VAFB SLC-4E

Total payload for NASA (CRS): 45,596 kg

Avg. payload, F9 v1.1 boosters: 2,534.7 kg

Max payload carried: 15,600 kg (F9 B5)

Milestones

First successful ground-pad landing: 1 May 2017

2015 drone-ship landing failures: 2 (F9 v1.1 B1012, B1015 — CCAFS LC-40)

4,000–6,000 kg drone-ship successes: F9 FT B1021.2, B1022, B1026, B1031.2

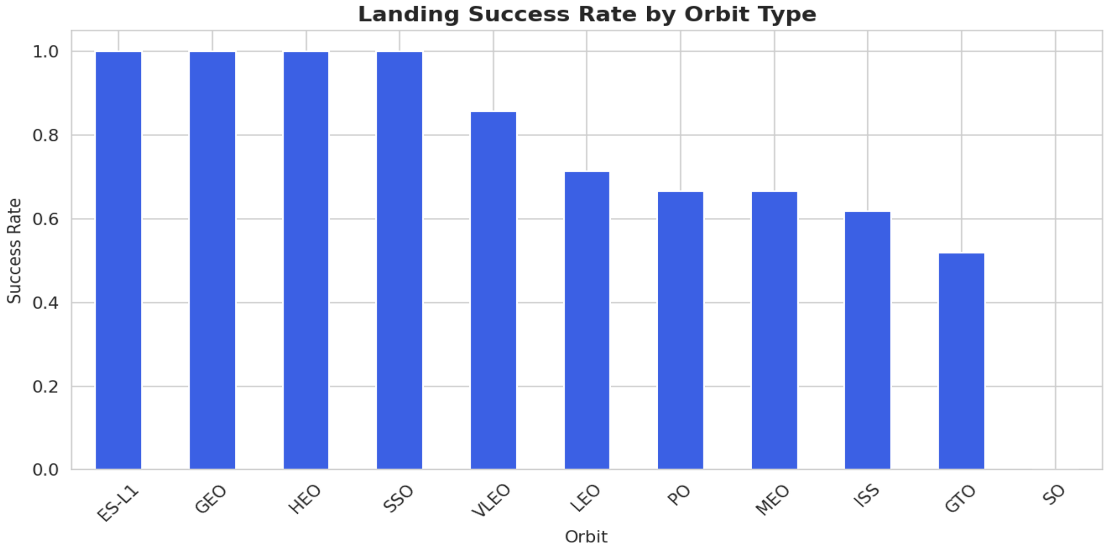
These figures ground the EDA charts in exact, query-verified numbers rather than visual estimates.

EDA SQL Results — Success Rates & Rankings

Landing outcome ranking, 2010–2017 (by count):

Landing Outcome	Count
Success	20
No attempt	10
Success (drone ship)	8
Success (ground pad)	6
Failure (drone ship)	4
Failure	3
Controlled (ocean)	3
Failure (parachute)	2

Mission outcome (2010–2020): 98 of 101 recorded outcomes were successful missions — first-stage recovery success is the harder, separate problem this model targets.

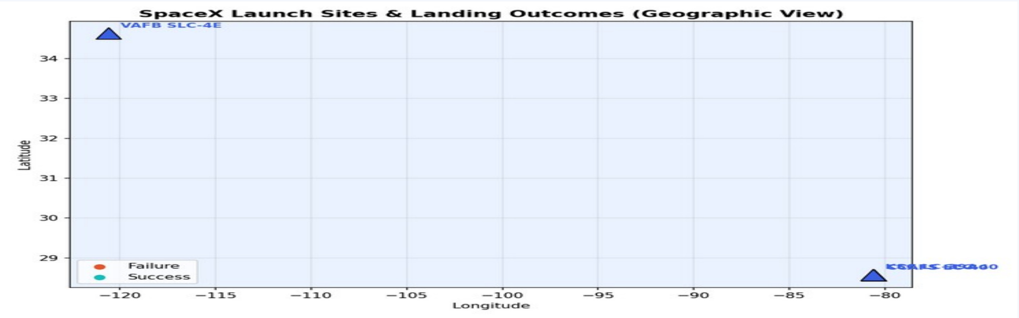


Interactive Visual Analytics

Two interactive tools let stakeholders explore the data themselves rather than reading static charts:

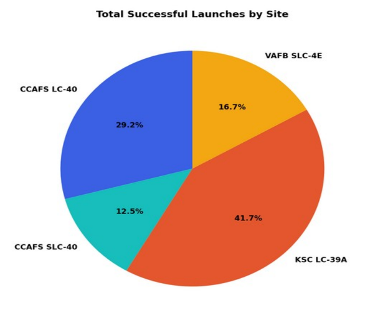
Folium Map

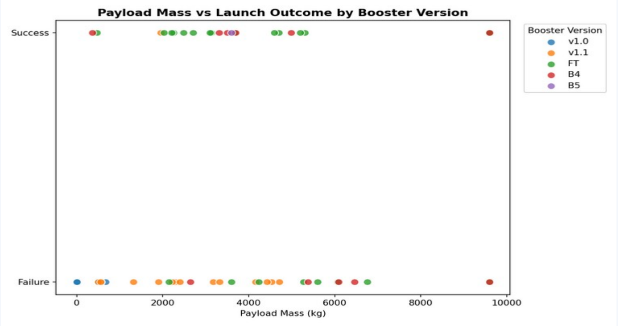
Launch-site markers, per-launch outcome markers (color-coded), and proximity distance lines (coastline, city, rail).



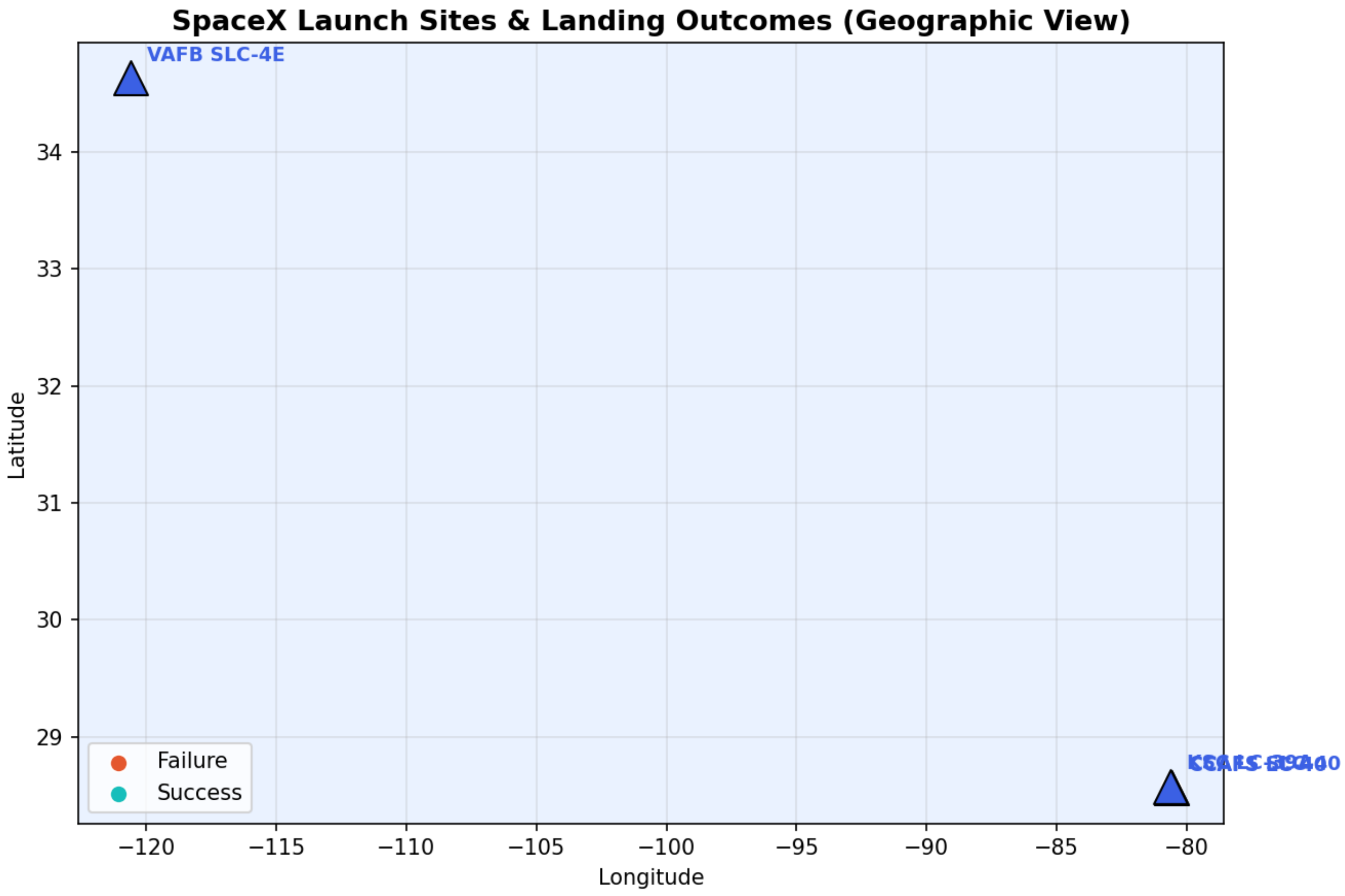
Plotly Dash App

Dropdown to filter by launch site; live-updating success-rate pie chart and payload-vs-outcome scatter plot.





Launch Site Markers & Launch Records



All 4 launch sites plotted with markers.
Every individual launch (55 records) plotted as a marker cluster, colored green for a successful landing and red for a failed/no-attempt landing — letting you see which sites cluster more successes.

Observation: KSC LC-39A and CCAFS SLC-40 show visibly denser green clusters than VAFB SLC-4E, consistent with their higher launch-cadence, more-refined-process profile.

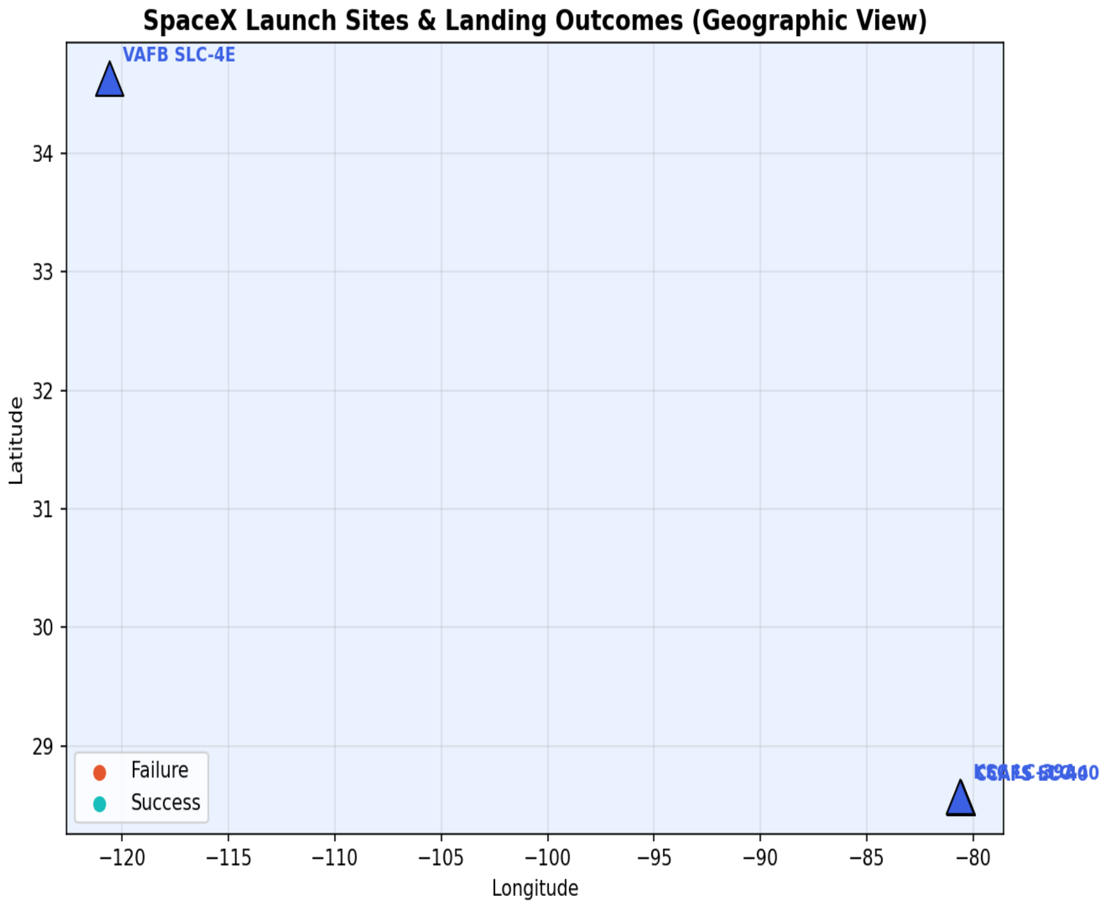
GitHub: https://github.com/YOUR-USERNAME/spacex-falcon9-capstone/blob/main/notebooks/06_folium_map.py

Proximity Analysis

Using the haversine formula, we measured distance from KSC LC-39A to nearby

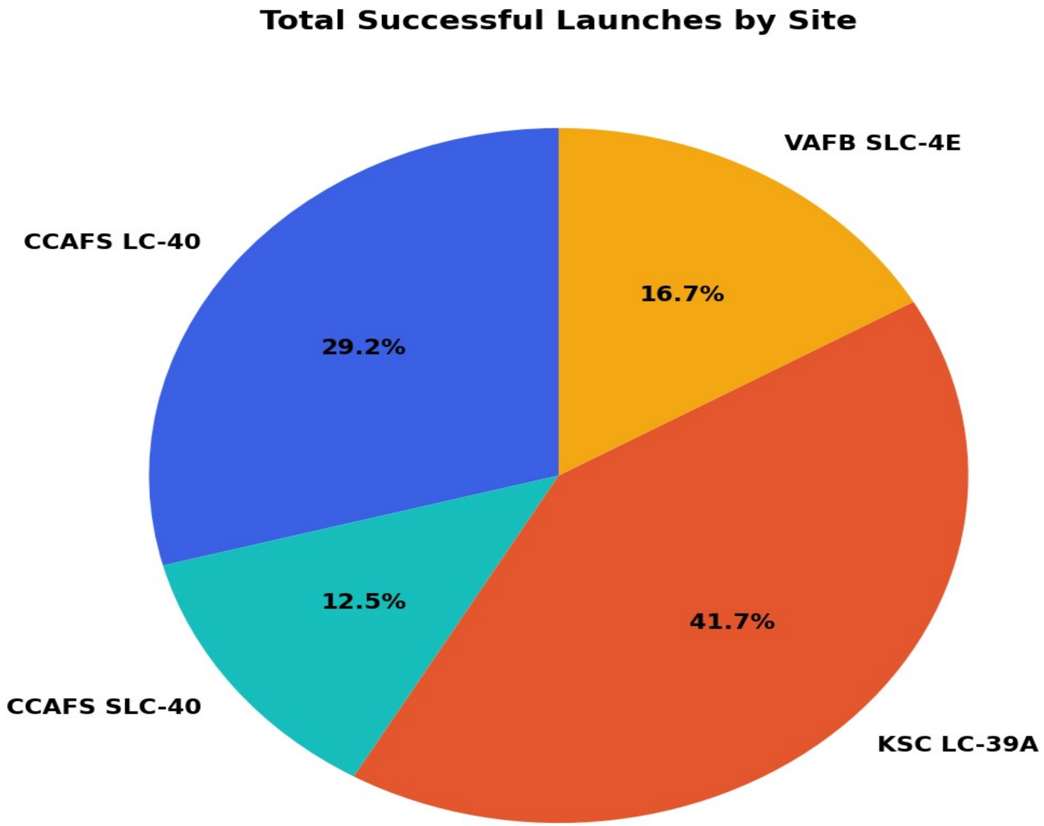
Feature	Distance from Launch Site
Nearest coastline point	7.43 km
Nearest city (Titusville area)	20.50 km

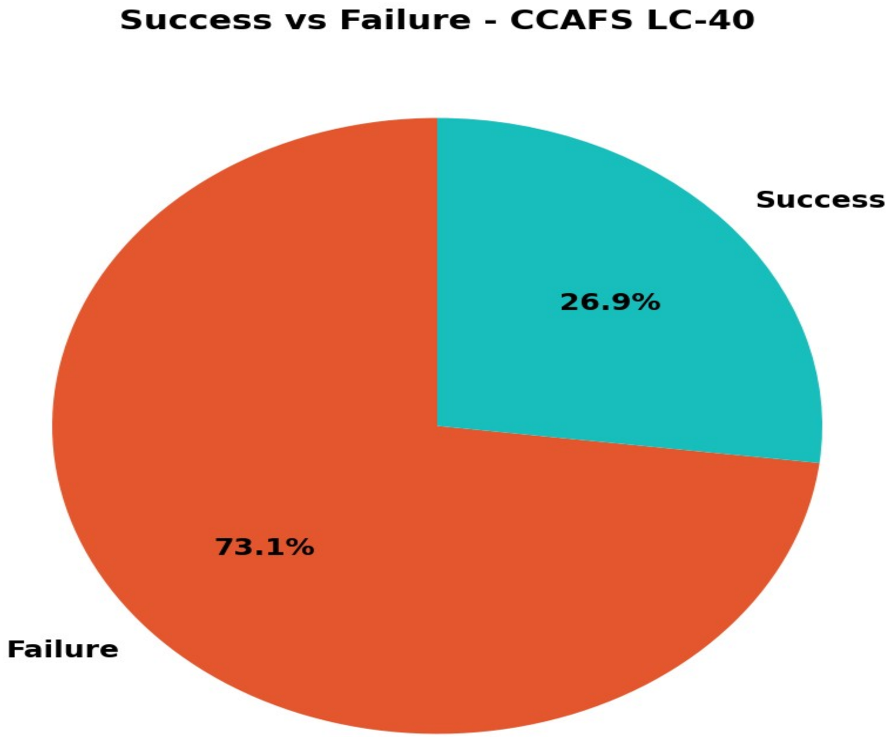
Why this matters: Launch sites are deliberately sited close to coastlines (for downrange safety over open water) but a safe distance from population centers and rail lines — a real siting constraint any competing launch provider would need to replicate.



GitHub: https://github.com/YOUR-USERNAME/spacex-falcon9-capstone/blob/main/notebooks/06_folium_map.py

Success Rate — Pie Charts

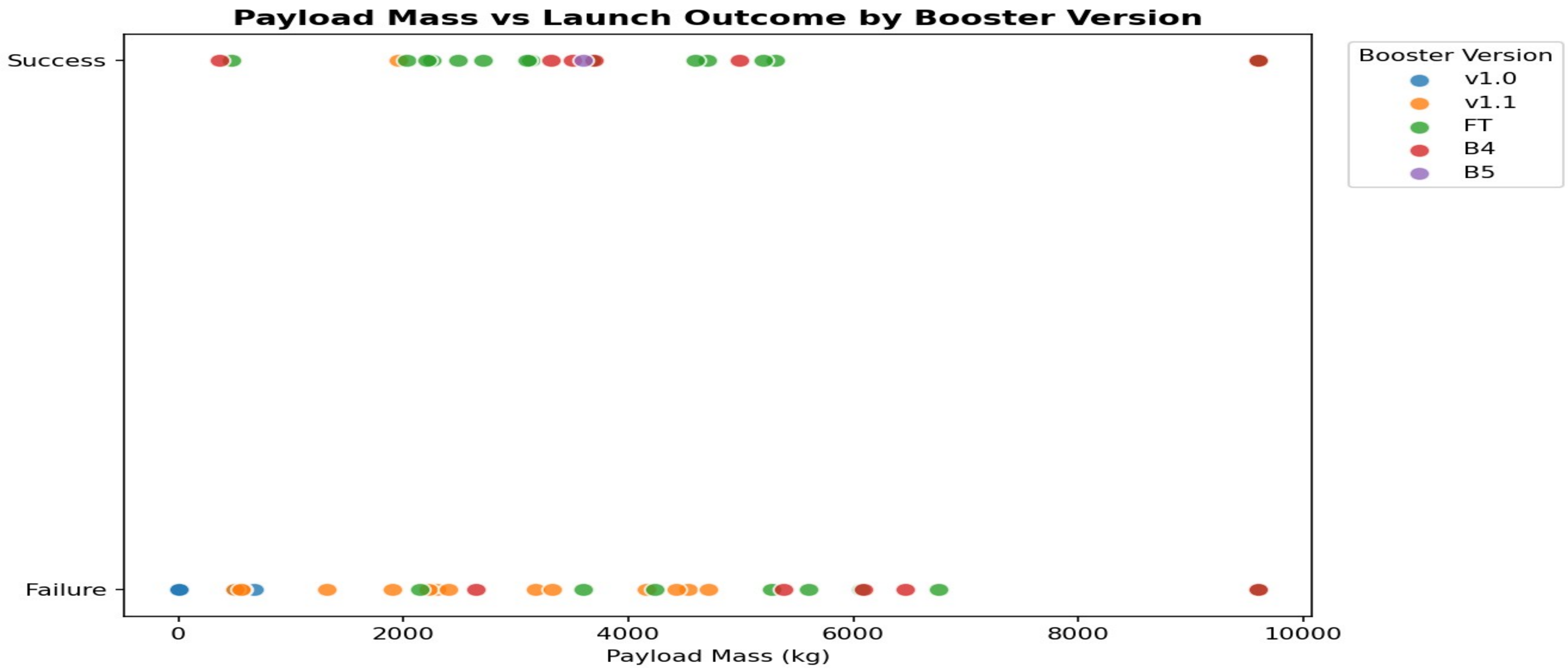




Left: share of total successful landings contributed by each site. Right: success-vs-failure split at the top-volume site (CCAFS LC-40) — filterable live via the dashboard dropdown.

GitHub: https://github.com/YOUR-USERNAME/spacex-falcon9-capstone/blob/main/dashboard/app.py

Payload Mass vs. Launch Outcome



A RangeSlider (0–10,000 kg) lets users narrow the payload window; the scatter plot recolors by booster version category (v1.0 / v1.1 / FT / B4 / B5), revealing that newer booster generations succeed reliably across a much wider payload range than early versions.

GitHub: https://github.com/YOUR-USERNAME/spacex-falcon9-capstone/blob/main/dashboard/app.py

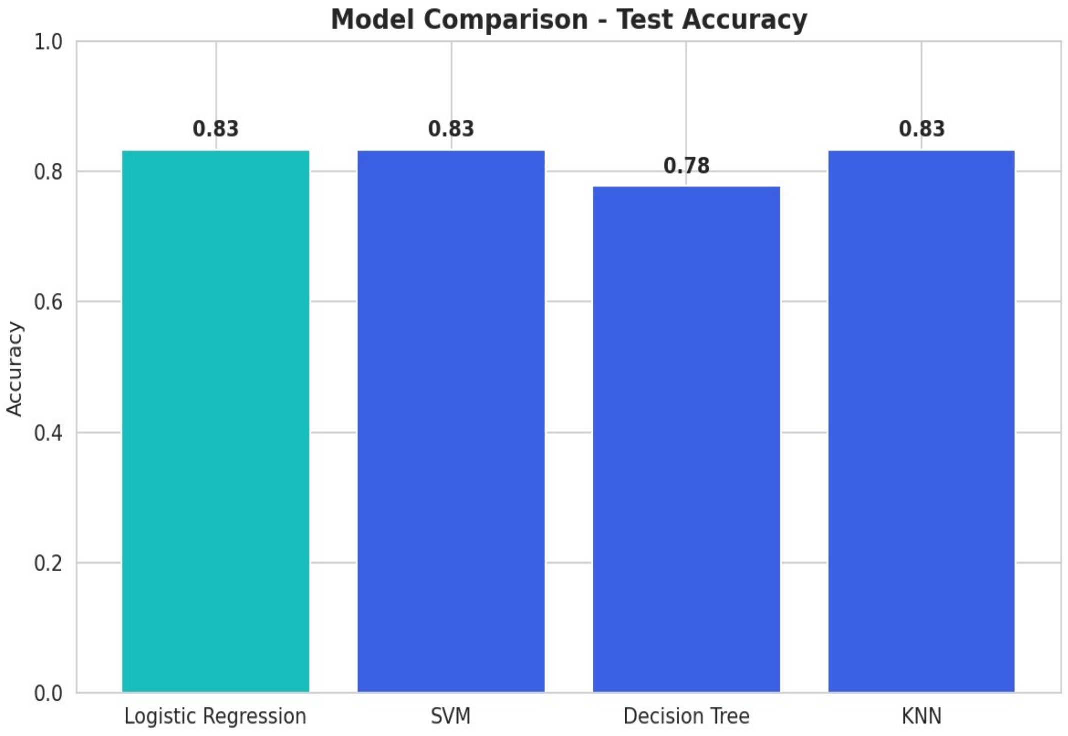
Model Development

Features: 80 one-hot-encoded columns (orbit, launch site, landing pad, booster serial) plus flight number, payload mass, flights, grid fins, reused, legs, and block — standardized with StandardScaler.

Split: 80% train / 20% test (72 / 18 launches), stratified via random_state=2.

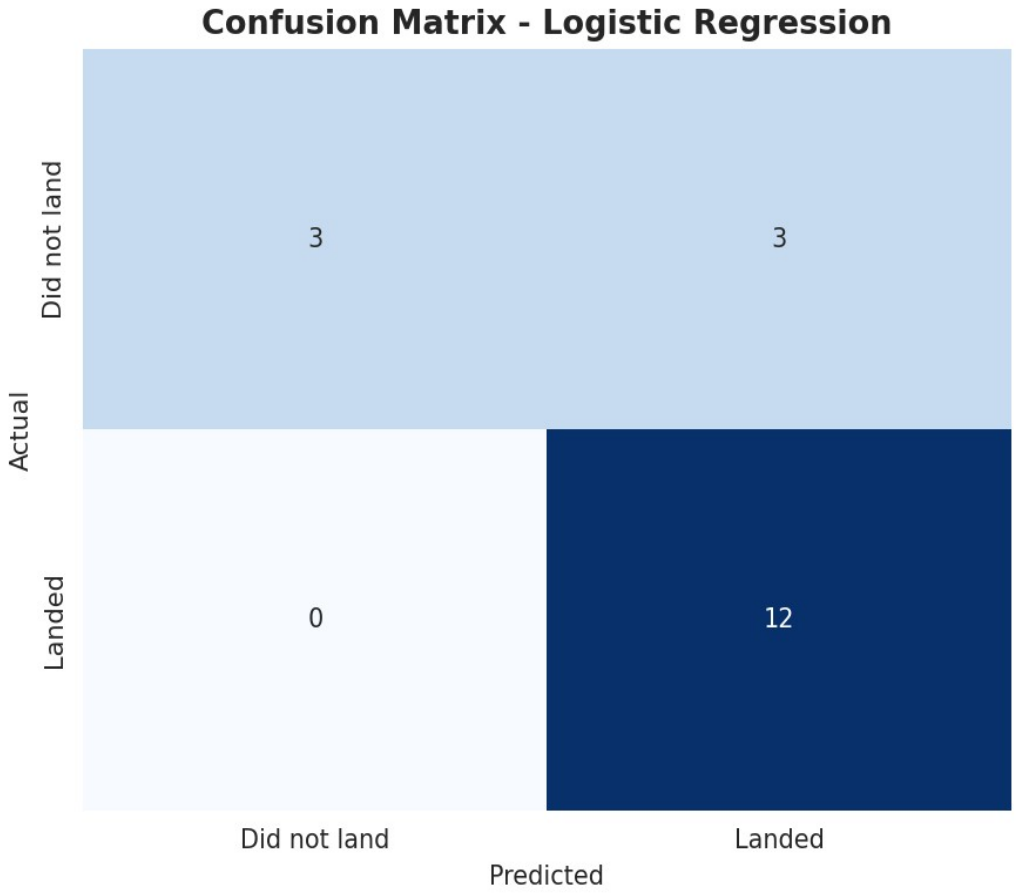
Models compared (each tuned via 10-fold GridSearchCV):

- Logistic Regression — C, penalty, solver
- Support Vector Machine — kernel, C, gamma
- Decision Tree — criterion, max_depth, min_samples_split
- K-Nearest Neighbors — n_neighbors, distance metric (p)



All four models converged near 83% test accuracy, with Decision Tree slightly behind — signal that the engineered features are strong and model choice matters less than feature quality here.

Evaluation Results & Confusion Matrix



Model	CV Score	Test Accuracy
Logistic Regression	0.821	0.833
SVM	0.834	0.833
KNN	0.834	0.833
Decision Tree	0.804	0.778

Reading the confusion matrix (18 test launches): 12 successful landings correctly predicted, 3 unsuccessful landings correctly predicted, 3 unsuccessful landings misclassified as successful, 0 successful landings missed. The model never misses a true landing — it's conservative in the safer direction for cost bidding.

Best Model & Conclusion

Best Model: Logistic Regression

Best params: C=1, L2 penalty, lbfgs solver

Why it wins: Ties SVM and KNN on raw accuracy (83.3%) but is far more interpretable — each feature's coefficient directly shows its effect on landing-success odds, which matters for explaining a cost bid to stakeholders. It's also the cheapest to retrain as new launches arrive.

Conclusion & Insights

- Landing success is highly learnable from launch parameters — 83% test accuracy from a simple, interpretable model.
- Orbit type and payload mass are the strongest visual predictors; heavier GTO missions land less reliably than light LEO/ISS missions.
- Success rate improved dramatically over time (2013 → 2020), reflecting SpaceX's iterative process improvements — not a static property of the hardware.
- Recommendation: a competing bidder can use this model as a first-pass cost-risk estimator, refining it further with more recent launches and continuous re-training.